

Untitled

2026-06-14

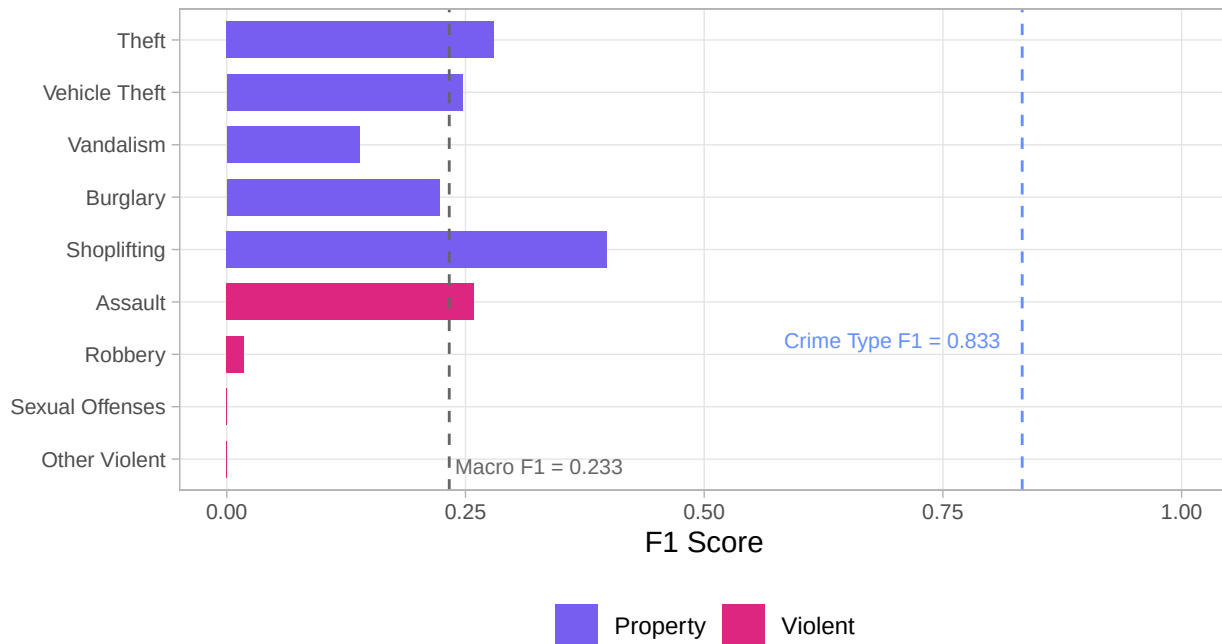
```
xgb_f1_per_class <- c(
  "Theft"          = 0.280,
  "Vehicle Theft"  = 0.247,
  "Vandalism"      = 0.139,
  "Burglary"       = 0.223,
  "Shoplifting"    = 0.398,
  "Assault"        = 0.259,
  "Robbery"        = 0.018,
  "Sexual Offenses" = 0.000,
  "Other Violent"  = 0.000
)

xgb_macro_f1 <- 0.233
xgb_f1_type <- 0.833

tibble(
  Subtype = factor(subtype_levels, levels = rev(subtype_levels)),
  Type    = c(rep("Property", 5), rep("Violent", 4)),
  F1      = xgb_f1_per_class[subtype_levels]
) |>
  ggplot(aes(x = Subtype, y = F1, fill = Type)) +
  geom_col(width = 0.7) +
  geom_hline(yintercept = xgb_macro_f1,
             linetype = "dashed", color = "gray40", linewidth = 0.5) +
  geom_hline(yintercept = xgb_f1_type,
             linetype = "dashed", color = ibm_blue, linewidth = 0.5) +
  annotate("text", x = 0.6, y = xgb_macro_f1,
           label = paste0(" Macro F1 = ", round(xgb_macro_f1, 3)),
           vjust = -0.4, hjust = 0, size = 2.8, color = "gray40") +
  annotate("text", x = 3, y = xgb_f1_type,
           label = paste0("Crime Type F1 = ", round(xgb_f1_type, 3)),
           vjust = -0.4, hjust = 1.1, size = 2.8, color = ibm_blue) +
  scale_fill_manual(values = c("Property" = ibm_purple, "Violent" = ibm_magenta)) +
  scale_y_continuous(limits = c(0, 1)) +
  coord_flip() +
  labs(title = "XGBoost Per-Subtype F1 Score",
       subtitle = "Gray dashed = subtype macro F1; blue dashed = crime type F1",
       x = NULL, y = "F1 Score", fill = NULL) +
  theme_ibm() +
  theme(legend.position = "bottom")
```

XGBoost Per-Subtype F1 Score

Gray dashed = subtype macro F1; blue dashed = crime type F1

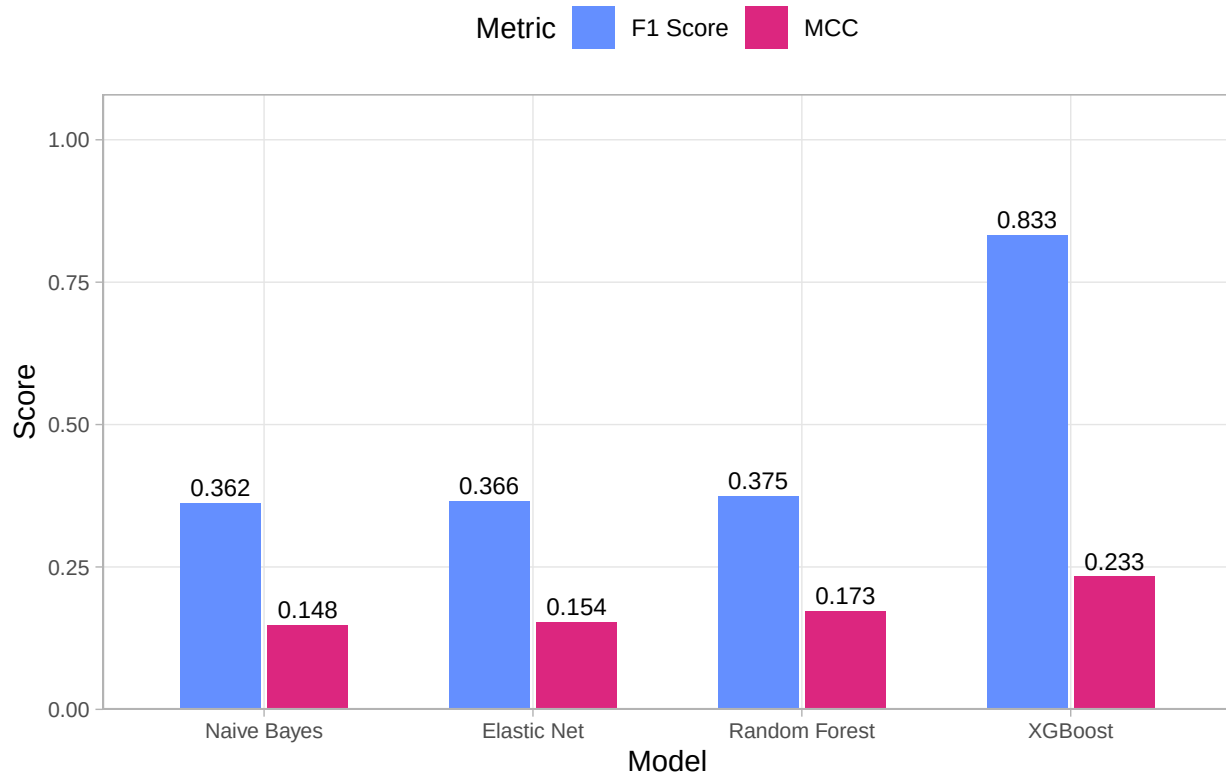


```
##f1 and mcc comparison across all models - replace values with actual results
model_metrics <- tibble(
  model = factor(rep(c("Naive Bayes", "Elastic Net", "Random Forest", "XGBoost"),
                    each = 2),
                levels = c("Naive Bayes", "Elastic Net", "Random Forest", "XGBoost")),
  metric = rep(c("F1 Score", "MCC"), times = 4),
  value = c(
    0.362, 0.148, ## naive bayes
    0.366, 0.154, ## elastic net
    0.375, 0.173, ## random forest
    0.833, 0.233  ## xgboost
  )
)

ggplot(model_metrics, aes(x = model, y = value, fill = metric)) +
  geom_col(position = position_dodge(width = 0.65), width = 0.6) +
  geom_text(
    aes(label = sprintf("%.3f", value)),
    position = position_dodge(width = 0.65),
    vjust = -0.4, size = 3
  ) +
  scale_fill_manual(
    values = c("F1 Score" = ibm_blue, "MCC" = ibm_magenta),
    name = "Metric"
  ) +
  scale_y_continuous(limits = c(0, 1), expand = expansion(mult = c(0, 0.08))) +
  labs(
    title = "Crime Type Model Performance: F1 Score and MCC",
    x = "Model",
    y = "Score"
  )
```

```
) +
theme_ibm() +
theme(legend.position = "top")
```

Crime Type Model Performance: F1 Score and MCC



```
##f1 and mcc comparison across all models - replace values with actual results
model_metrics <- tibble(
  model = factor(rep(c("Naive Bayes", "Elastic Net", "Random Forest", "XGBoost"),
    each = 2),
    levels = c("Naive Bayes", "Elastic Net", "Random Forest", "XGBoost")),
  metric = rep(c("F1 Score", "MCC"), times = 4),
  value = c(
    0.228, 0.089, ## naive bayes
    0.057, 0.028, ## elastic net
    0.156, 0.094, ## random forest
    0.233, 0.121 ## xgboost
  )
)

ggplot(model_metrics, aes(x = model, y = value, fill = metric)) +
  geom_col(position = position_dodge(width = 0.65), width = 0.6) +
  geom_text(
    aes(label = round(value, 3),
      position = position_dodge(width = 0.65),
      vjust = -0.4, size = 3)
  ) +
  scale_fill_manual(
    values = c("F1 Score" = ibm_blue, "MCC" = ibm_magenta),
```

```

    name = "Metric"
  ) +
  scale_y_continuous(limits = c(0, 1), expand = expansion(mult = c(0, 0.08))) +
  labs(
    title = "Crime Subtype Model Performance: F1 Score and MCC",
    x = "Model",
    y = "Score"
  ) +
  theme_ibm() +
  theme(legend.position = "top")

```

Crime Subtype Model Performance: F1 Score and MCC

